

DATA. INSIGHT. PERFORMANCE.

FOOTBALL ANALYTICS DATABASE SYSTEM

A relational database solution for football performance analysis, scouting, and tactical decision making.



CASE STUDY: FC VIKTORIA 1889 BERLIN



RELATIONAL DATABASE DESIGN
Normalized schema with 8 core tables, primary & foreign keys, and ERD modelling.



ADVANCED ANALYTICS
Goal contribution, fitness classification, passing analysis, and transfer watchlist.

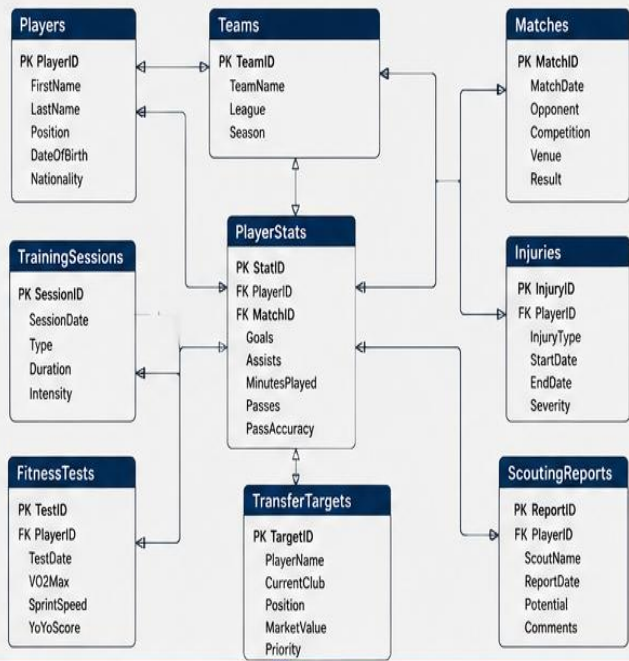


SQL IMPLEMENTATION
Views, stored procedures, CASE expressions, indexing and query optimization.



INTERACTIVE DASHBOARDS
Power BI dashboards for performance monitoring and tactical insights.

DATABASE DESIGN (ERD)



POWER BI DASHBOARD PREVIEW



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INTRODUCTION

With football being a core value most of my life, I can confidently say that the sport has experienced a shift in recent years in the sense that it is slowly moving beyond the traditional coaching where decisions were solely made based on intuitions. Clubs these days rely on analysis systems to improve performance on the pitch. Sportmonks (2026) argues that analyzing match and training session's data allows coaching staff to make decisions based on facts. According to Hudl (2018), German football club SC Paderborn 07 used Hudl and Sportscore which are video analysis platforms for match preparation and developing youth players. Coaching staff say they used statistical analysis to study opponents and improve player's performance. The club says it is important to have a centralized database for monitoring player development for both the first team and youth team (Hudl, 2018). With teams now relying heavily on wearable GPS to support on the pitch performances, data is almost always being created and managed throughout a football season (Sportmonks, 2026). FifPro says teams now collect player data for coaching and scouting. The FifPro article highlights that this new era of data driven football analysis means there is the need for centralized systems that help with managing football data more effectively.

For teams to manage football data, they require a relational database system. Gillenson (2011) explains that relational databases remove any form of data redundancy by organizing data into domains that are connected through primary keys and foreign keys. Without a structured system, football data will be difficult to manage and makes it less useful for decision making.

Information from these database management systems help teams to gain a competitive advantage over opponents. The use of Power BI further support this process by creating visual reports that are easier to understand among coaches and players. Post (2004) notes that using database applications within an organization improves decision making by allowing users to analyze data more efficiently. In modern football, quick access to accurate data can influence matches in terms of team selection and tactics. For example; analyzing data on penalty kicks

from opponents' previous penalties is often studied to identify shooting patterns like; preferred shooting direction, placement, and technique. In the 2024 UEFA Euros, goalkeeper Jordan Pickford was seen with a water bottle containing information on the penalty preferences of Swiss players. Instead of guessing, he used past data to support where to dive for each penalty taker, which aided his decision when saving Manuel Akanji's penalty kick to secure a semi final spot for England. (Sky News, 2024). Figure 1 provides a visual representation.



Figure 1: Example of how Data Analysis is being used in Modern Football

Since arriving in Germany, I have closely followed FC Viktoria 1889 Berlin's men's team and eventually got to understand the playing style and football operations in the club. Because of this familiarity, the club was used as a case study for designing a football relational database to show how the system can support analysis and decision making in football. The system is designed to manage player records, match statistics, training loads, injury reports, scouting information, and transfer decisions. The system is to be used by coaches, analysts, medical staff, and club management for analysis and decision making.

MySQL and Microsoft Power BI were tools used for setting up and designing this system. This study demonstrates how relational database systems can support decision making in football.

METHODOLOGY

System Design

A relational database system was selected since it organizes data into related tables for easy access (Powell, 2006). According to Bagui & Earp (2011), setting up a database starts with a conceptual design, also known as high abstract stage through to logical design also referred to as Medium abstract stage and physical design or low abstract stage. The focus of this RDBMS was on building a structured system for managing football related data to be used for analytics and decision making within FC Viktoria Berlin.

Forward engineering guides were carefully followed in developing this system (Bagui & Earp, 2011; Ramakrishnan, 2014). They include:

1. Understand what data needs management at FC Viktoria Berlin.
2. Create domains based on understanding of data to be used.
3. Create Tables
4. Define keys to create relationships between tables.
5. In all domains, ensure data is consistent
6. Create a RDBMS using My SQL Workbench.
7. Insert sample data in relation to the domains.
8. Run queries to support analysis in the database.
9. Improve quality of long queries.
10. Validate outputs.
11. Use graphs to communicate results.

The design process started with identifying football data worth managing in the club. These activities included player management, match analysis, injury monitoring, training load tracking, and transfer scouting operations. The relational database system then follows a structured process consisting of database modelling, normalization, SQL implementation, query

improvement, validation, and business intelligence visualization. The database system stores football data as well as to support decision making through analytical queries and visual dashboards (Bagui & Earp, 2011).

CAP THEOREM

Since football data is stored in databases shared across connected systems of different domains, as a data engineer I have to decide which system properties should have priority. These properties include; data accuracy, system availability or partition tolerance. Eric Brewer argues that a distributed database system cannot have Consistency, Availability, and Partition Tolerance properties altogether. Instead, a system can only prioritize two of these properties at any given time (Rahimi & Haug, 2015). This theory is called the CAP Theorem.

Consistency in the CAP theorem means if a player's or match data are updated, the latest version can be accessed on all servers. (Rahimi & Haug, 2015).

If the system is capable of responding to user requests even when network problems exist, this system has a property termed as Availability in the CAP Theorem. For example, analysts should still have access to player data even if one server becomes unavailable (Rahimi & Haug, 2015).

Partition Tolerance allows the system to remain working when communication between servers is disrupted (Rahimi & Haug, 2015; GeeksforGeeks, 2024). If this database is available in offices in Neukolln, Kreuzburg and Wittenau with each office managing different data domains communicating through the internet, and Neukolln loses internet connection to the other two offices, this is called a network partition. Partition Tolerance in such a case means the system can continue operating regardless of this challenge.

The ideal priorities to be considered for this study are Availability and Partition Tolerance because analysts and coaches need continuous access to data during matches and training. However, data on player medical records will only make sense to prioritize Consistency to avoid accessing outdated data.

Database Relationships

Based on the domains: player management, performance analysis and fitness, the main entities used for the database system included Players, Matches, Player Stats, Injuries, Training Load, Transfer Targets, Transfer Decisions and Clubs.

In order to retrieve data easily, primary keys and foreign keys were used to create relationships across domains (Gillenson, 2011). In terms of domain relationships, one-to-many relationships exist between Players and Training Load, Players and Injuries, as well as Players and Transfer Decisions tables. Also, since a player can play in multiple matches and each match is played by multiple players, a many-to-many relationship was created between players and matches tables. To support this relationship, a new junction table named player stats was created to link both tables. This table consists of player stats such as goals and assists for each player during matches.

After further observations, an as-is to to-be analysis was performed on the transfer targets table. Club information kept on repeating in the table causing redundancy. To address this issue, normalization took place. Team names were transferred into a separate table named clubs. Instead of storing club names repeatedly for every transfer target, club ID was used as a foreign key to reference the clubs table directly. This improved overall consistency within the player recruitment domain (Ramakrishnan, 2014).

The following describes the keys used in each table domain list;

Players - Player ID is a primary key used for identifying each player in FC Viktoria Berlin. No no foreign key exists in this table

Matches - Match ID is a primary key used to identify each match played, also no foreign key was created for this table.

Injuries - Injury ID is a primary key used to identify each injury diagnosis, while Player ID is a foreign key that connects injury records to the Players table.

Training load - Load ID is a primary key used for identifying each training load record, and Player ID is a foreign Key that identifies each player's training load.

Transfer Decisions - Recommendation ID is a primary key identifying a player's current transfer status, while Player ID is a foreign key connecting transfer decisions to each player.

Transfer Targets - Target ID is the primary key used to identify each player on the teams transfer radar to recruit, while Club ID is a foreign key connecting with the Clubs table (which contains club information of other clubs).

Clubs - Club ID is a primary key identifying each club for which transfer targeted players belong to.

Player Stats - Player ID and Match ID are both primary and foreign keys, referencing Players and Match tables respectively.

This is further illustrated in Figure 2

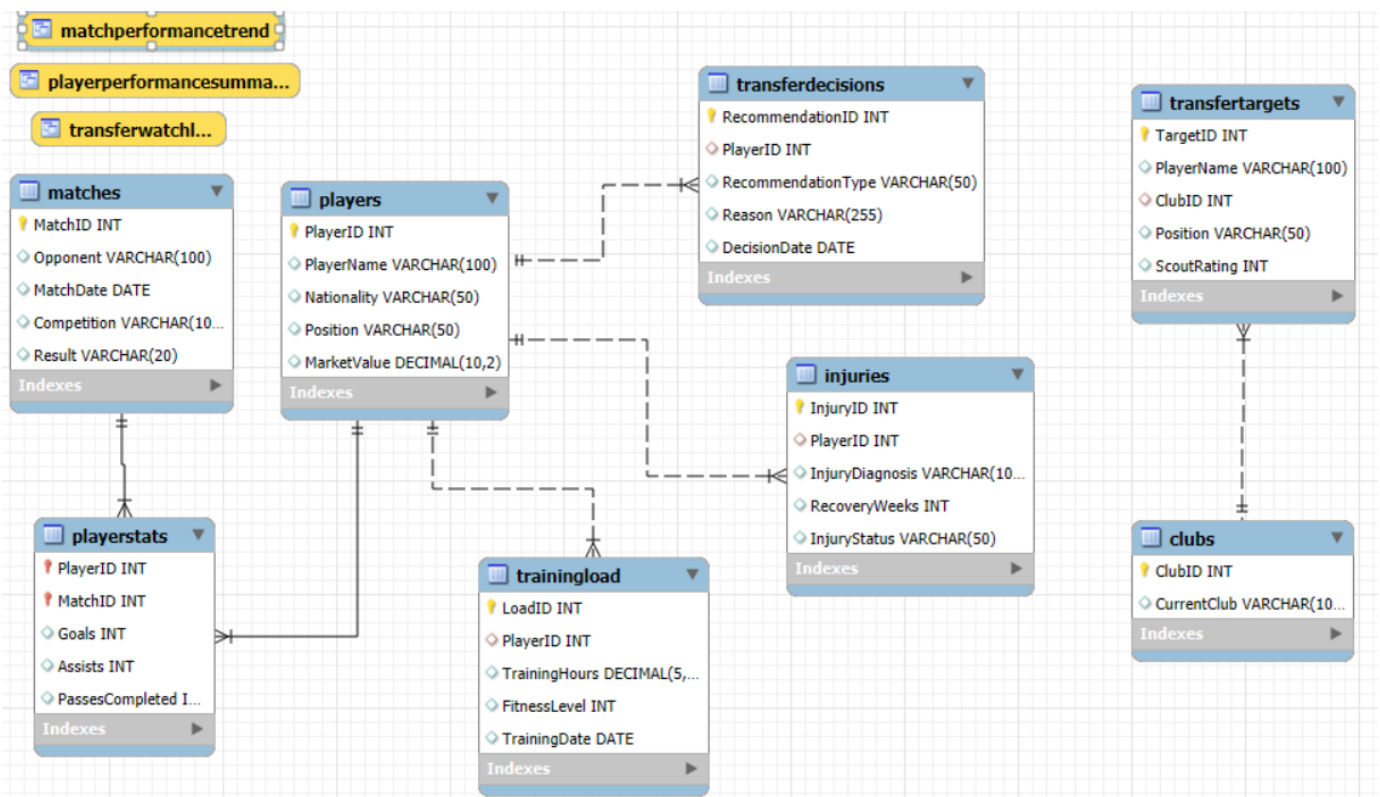
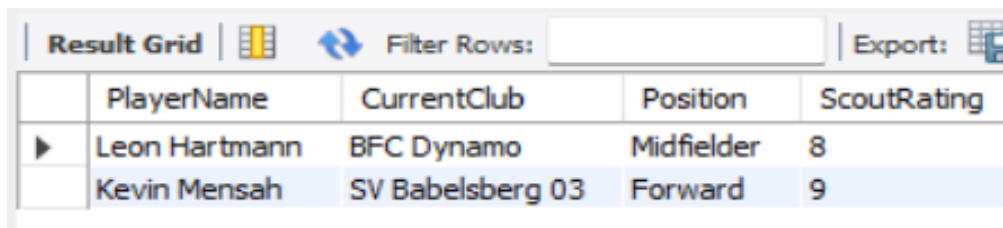


Figure 2: Database ERD

Business Logic

In carrying out FC Viktoria 1889 Berlin's relational database in MySQL, sample data was used to imitate real world football operations. Based on the inserts, SQL queries were executed to support football analytics. These queries focused on identifying players with the highest goal contributions, players with most completed passes in matches and monitoring player fitness conditions. The system includes business logic to help with decision making on and off the pitch. For example, player fitness levels were clustered into "Fit", "Moderate Fitness", and "Needs Recovery" groups using SQL CASE expressions. This allows for easy understanding of numerical fitness (Post, 2004). In addition, goal contributions were calculated by combining goals and assists using JOIN and GROUP BY to identify players with attacking returns. Finally, SQL views were created to allow the reuse of queries for easier analysis (Ramakrishnan, 2014). A Transfer Watchlist view was created to assist management in identifying future player recruitment. This was done by combining data from the Transfer Targets and Clubs tables to return transfer targets with scout ratings greater than 7 as shown in Figure 3.



The image shows a screenshot of a database application's 'Result Grid'. At the top, there is a toolbar with a 'Filter Rows' input field and an 'Export' button. Below the toolbar is a table with the following data:

	PlayerName	CurrentClub	Position	ScoutRating
▶	Leon Hartmann	BFC Dynamo	Midfielder	8
	Kevin Mensah	SV Babelsberg 03	Forward	9

Figure 3: Results from transfer watch list view

Stored Procedure Implementation

A stored procedure was created on goal contributions and executed using the Call command. Instead of repeatedly executing long queries, this stored procedure allows for query reusability which saves time (Coronel & Morris, 2015).

Query Optimization

The goal contribution query used large JOIN operations which gradually became slower with more records in the system. Krantz and Jonker (n.d.) state that in optimizing a query, the focus should be on improving how fast queries should produce an output by reducing table scan delays, allowing faster retrieval of information. With this in mind, the query was optimized by using an index. It was created on the player id attribute within the player statistics table to speed up data retrieval.

System Validation

Tables, data formats and relationships were inspected after implementation to validate the system as a whole. In addition, SQL query results were further visualized using Microsoft Power BI dashboards test whether the system is capable of communicating valid results to support football operations (Post, 2004). This creates a link between back-end and front-end. We see this type of visual reporting being used in modern football by coaches and analysts during tactical preparation (Sumpter, 2016).

FINDINGS & ANALYSIS

This chapter shows results from queries can generate important information on player performance, fitness and possession and how the overall database can improve performance within the team. These findings proved how relational databases and business intelligence tools can support decision making in football.

Goal Contribution Analysis

Identifying players with the highest goal scoring contributions within the squad is a priority for this system. As illustrated in Figure 4, the query combines Players and Player Stats tables using a JOIN operation and calculates total goals and assists. The query also includes overall goal contribution for each player, calculated by summing goals and assists.

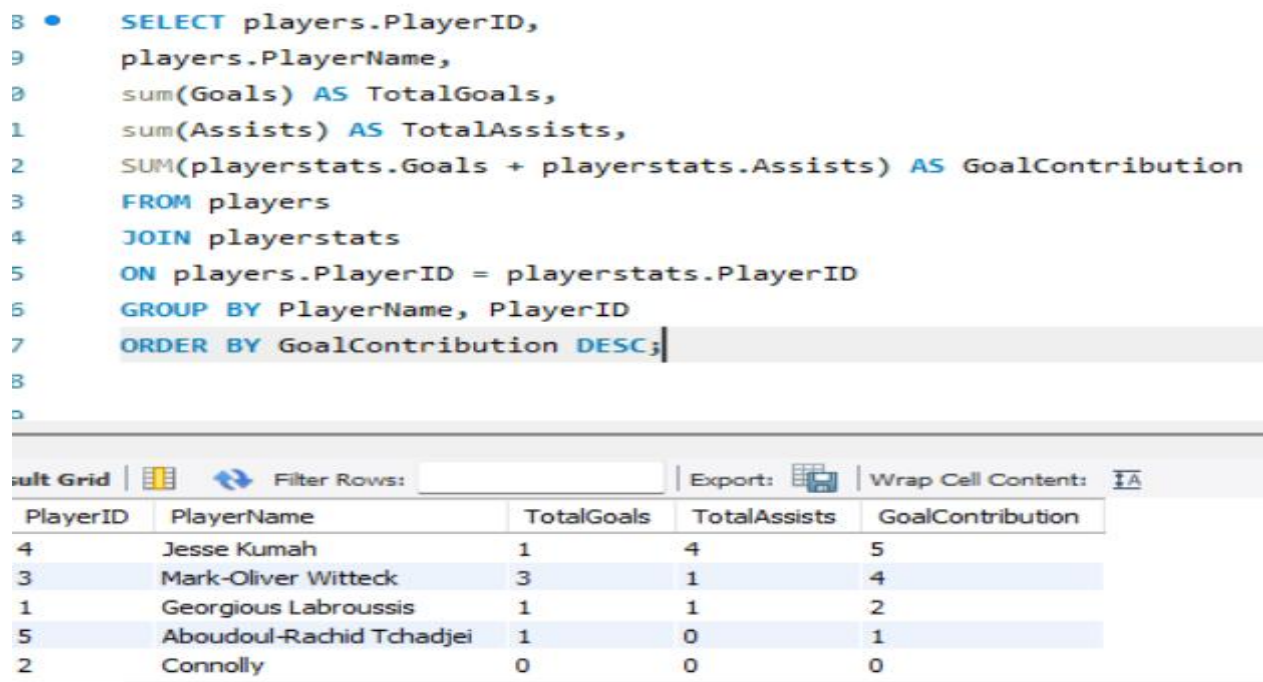


Figure 4: SQL query and result for player goal contribution

Based on the results, it is seen that players with high goal contribution values contribute most to the team's progress. This information can assist coaching staff in modern football when selecting players based on a tactical system (Sportmonks, 2026). As shown in figure 5, a clustered bar chart was created in Power BI to visualize total goal contributions for each player. Such a visual improves understanding by transforming query outputs into more understandable information for coaches, analysts, and players, to support better decision making (Sumpter, 2016).



Figure 5: Bar chart showing goal contributions among players

Player Fitness Analysis

Mahammad (2024) argues fitness is important in football because it influences player performance and reduces the risk of injuries. To evaluate a player's readiness level, a CASE statement was used to classify players according to their recorded fitness levels. This query groups players in: "Fit", "Moderate Fitness", and "Needs Recovery".

- ```
SELECT players.PlayerID, PlayerName, players.Position , Fitnesslevel,
CASE
 WHEN FitnessLevel > 7 THEN 'Fit'
 WHEN FitnessLevel BETWEEN 6 AND 7 THEN 'Moderate Fitness'
 ELSE 'Needs Recovery'
END AS FitnessStatus
FROM players
JOIN trainingload
ON players.PlayerID = trainingload.PlayerID;
```

Figure 6: Grouping fitness levels

Query results from Figure 6 returned player fitness conditions within the squad. Players classified as “Fit” are prepared to play any match possible, while “Moderate Fitness” may play matches but need limited playing time whereas players under “Needs Recovery” need injury rehabilitation and cannot participate in games. The CASE expression simply converts the numerical fitness data into understandable categories. Such analysis can support team selection decisions and workload management for players (Mahammad, 2024).

A donut chart and a table visual was created in Power BI to show results of the three fitness categories as illustrated in figure 7. It provides the coaching and medical staff a percentage of fit players going into a match week.

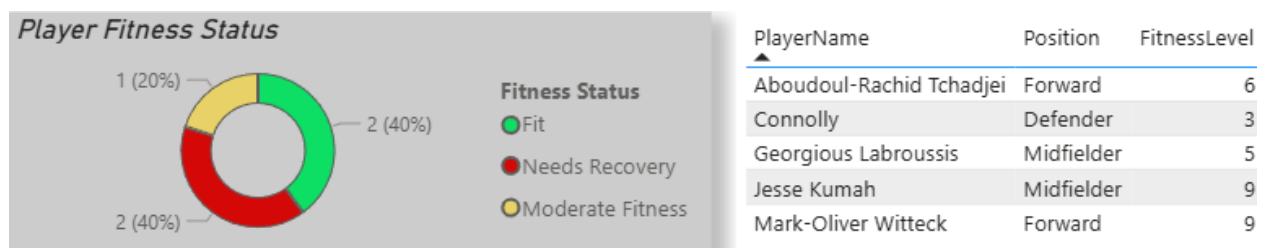


Figure 7: Visual Communicating Fitness Levels

## Possession Football and Passing Contribution Analysis

After watching a few games, it was clear that FC Viktoria Berlin is a possession based team that likes to control games. Research shows that the more a team keeps possession of the ball, the higher chance they stand in succeeding in domestic football leagues (Barça Innovation Hub, 2024).

```
SELECT
 p.PlayerName,
 p.Position,
 SUM(ps.PassesCompleted) AS TotalPasses,
 RANK() OVER (
 ORDER BY SUM(ps.PassesCompleted) DESC
) AS PassRank
FROM Players p
JOIN PlayerStats ps
ON p.PlayerID = ps.PlayerID
GROUP BY p.PlayerName, p.Position;
```

Figure 8: Query Ranking Players based on Total Passes Completed

Figure 8 shows a query that calculates and ranks players in FC Viktoria based on passes each player has registered. The results of this provides coaches with an idea of players that could suit a possession based tactical system. Based on the results, midfielders completed the most passes since they are known for linking defensive and attacking phases of play. Defenders with high completed passes are those coaches can consider if they want to start building up from the back as we see in the game play of Pep Guardiola's Manchester City side.

In Figure 9, a horizontal bar chart visibly ranks players according to completed passes to assist coaches in decision making by identifying players responsible for maintaining possession and moving the ball effectively during matches.

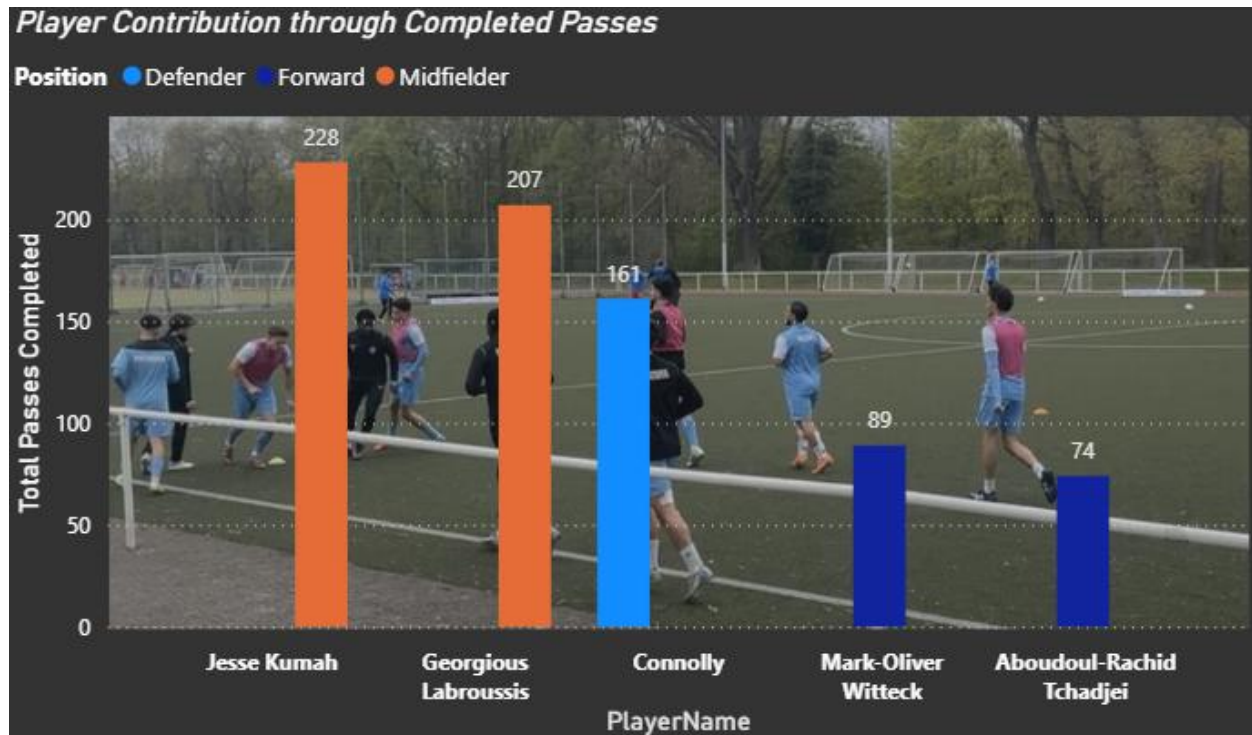


Figure 9: Bar Chart distribution showing completed passes

### Query Optimization Findings

As stated in the methodology, the goal contribution query was optimized to improve query response time (Krantz & Jonker, n.d.). This was done by creating an index on the Player ID attribute in the Player Stats table as seen in the query below.

```
CREATE INDEX idx_playerstats_playerid
```

```
ON PlayerStats(PlayerID);
```

Query response time improvement is proven in figure 10 in the following chapter.

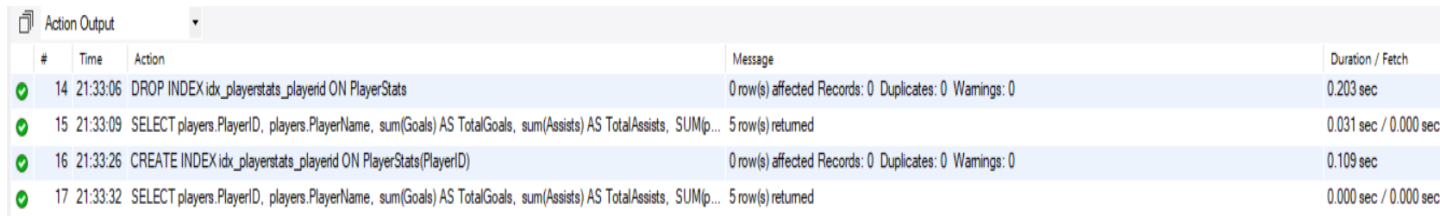


## SYSTEM EVALUATION

The ideal RDMS is evaluated using ACID principles to ensure data integrity (Cohen, 2026). Since this study uses a small quantity of sample data, although the ACID principles were considered, evaluation was centered more on removing redundancy, testing keys for data integrity, business logic and the systems ability to communicate results using Power BI

The use of primary and foreign keys helped with data consistency between related tables. Normalization solved redundancy in the recruitment domain by adding an extra table named 'Clubs' to separate club information from other data to make the database easier to maintain.

Adding an index was useful in the database because the execution time of the Goal Contributions query improved significantly from approximately 0.031 seconds to 0.000 seconds as shown in Figure 9. This demonstrates how indexing improves database performance. It shortens query response time to allow for faster analysis to be made in football where lots of data may be queried frequently for reporting and tactical analysis (Krantz & Jonker, n.d.).



| #    | Time     | Action                                                                                                        | Message                                                | Duration / Fetch      |
|------|----------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| ✓ 14 | 21:33:06 | DROP INDEX idx_playerstats_playerid ON PlayerStats                                                            | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.203 sec             |
| ✓ 15 | 21:33:09 | SELECT players.PlayerID, players.PlayerName, sum(Goals) AS TotalGoals, sum(Assists) AS TotalAssists, SUM(p... | 5 row(s) returned                                      | 0.031 sec / 0.000 sec |
| ✓ 16 | 21:33:26 | CREATE INDEX idx_playerstats_playerid ON PlayerStats(PlayerID)                                                | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.109 sec             |
| ✓ 17 | 21:33:32 | SELECT players.PlayerID, players.PlayerName, sum(Goals) AS TotalGoals, sum(Assists) AS TotalAssists, SUM(p... | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

Figure 10: Query Response Time after Indexing

SQL queries created useful information out of football data in the system. For example, fitness levels of players were automatically categorized into fitness groups with the help of CASE expression. This allows coaches and analysts to identify players who may need additional recovery or training support. The calculation of each player's goal contribution provided a measure for calculating how much value players add to winning matches.

Results from analysis were further communicated in the forms of graphs using Microsoft Power BI for easy understanding among team management.

Although the system achieved its objectives, there three major limitations;

1. The database was tested using sample data rather than live team data
2. Dataset size was relatively small compared to those used in professional football.
3. Predictive analysis should be made to further support decisions.

In the future, the system would use large amounts of real time data with predictive analytics to support advanced statistics such as expected goals (xG) to support decision making.

## CONCLUSION

Relational database management systems provide a structured means for data retrieval to be used for as making accurate decision based on facts instead of guessing. Insights can be created form all domains within the football environment to improve team performance (Sportmonks, 2026). According to Sportmonks, the more football clubs improve, the more they collect data from GPS tracking and performance monitoring platforms. Thus, the need to effectively manage club data.

This study on FC Viktoria 1889 Berlin's demonstrates how relational database systems and business intelligence tools can be used to manage football data across domains. Having followed the team closely, developing this system around the club helped combine personal interest with academic learning. With the use of queries, the system is capable of retrieving information to support decision making in the club. Queries such as Goal contributions, player fitness, and passing can be used for pre match analysis. The outputs of these queries were made easy to understand with the help of graphs created in Power BI.

Data collected in football allows teams such as FC Viktoria Berlin to have a competitive advantage over opponents by supporting accurate decision making (Hudl, 2018).

Overall, this study proves that RDBMS alongside BI tools can support football operations in modern day sports. Although the system was developed using sample data, it provides a baseline ideology of how this system can be implemented in a real world setting. With additional real time data and predictive analysis, this system can literally sky rocket the performances of FC Viktoria 1889 Berlin.

### Supporting Files

The complete SQL scripts used in developing the database is available here: [fc\\_viktoriaberlin\\_db.sql](#)

An overview of FC Viktoria Berlin's dashboard created in Power Bi is provided in figure 11 to communicate results from the RDBMS.



Figure 11: FC Viktoria Berlin dashboard

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